11.18 Solution: (a) In the rest frame of the particle, the electric and magnetic fields are given by

$$\mathbf{E}' = \frac{q\mathbf{r}'}{r'^3} = \frac{q\left(\mathbf{r}'_\perp + z'\hat{z}\right)}{r'^3}, \quad \mathbf{B}' = 0,$$

with the coordinates in the rest frame, which are related to those in the laboratory frame by

$$\mathbf{r}'_\perp = \mathbf{r}_\perp, \quad z' = \gamma(z - vt).$$

Then, using the coordinates in the laboratory frame, the fields are

$$\mathbf{E}' = \frac{q\left(\mathbf{r}_\perp + \gamma(z - vt)\hat{z}\right)}{[r_\perp^2 + \gamma^2(z - vt)^2]^{3/2}}, \quad \mathbf{B}' = 0.$$

Applying the Lorentz transformation, the fields in the laboratory frame are

$$\mathbf{E} = \frac{\gamma q\left(\mathbf{r}_\perp + \gamma(z - vt)\hat{z}\right)}{[r_\perp^2 + \gamma^2(z - vt)^2]^{3/2}}, \quad \mathbf{B} = \gamma\boldsymbol{\beta} \times \mathbf{E}' = \boldsymbol{\beta} \times \mathbf{E}.$$

The electric field can be expressed as

$$\mathbf{E} = \frac{q}{r_\perp^2}\left(\mathbf{r}_\perp + \gamma(z - vt)\hat{z}\right)\frac{\gamma/r_\perp}{\left[1 + \left(\dfrac{\gamma}{r_\perp}\right)^2 (z - vt)^2\right]^{3/2}}.$$

In the limit of $\beta \to 1$, we have $\gamma \to \infty$. In this limit, the electric field becomes singular. Consider the function

$$\varphi(z) = \frac{\alpha}{2(1 + \alpha^2 z^2)^{3/2}},$$

whose indefinite integral is

$$\Phi(z) = \frac{1}{2}\int \frac{dz}{(1 + \alpha^2 z^2)^{3/2}} = \frac{\alpha z}{2\sqrt{1 + \alpha^2 z^2}} + C.$$

We can verify that,

$$\int_{-\infty}^{+\infty} \varphi(z)dz = \lim_{z \to +\infty}\left(\frac{\alpha z}{2\sqrt{1 + \alpha^2 z^2}} - \frac{-\alpha z}{2\sqrt{1 + \alpha^2 z^2}}\right) = 1,$$

and

$$\int_{\varepsilon < |z| < +\infty} \varphi(z)dz = \lim_{z \to +\infty}\left(\frac{\alpha z}{\sqrt{1 + \alpha^2 z^2}} - \frac{\alpha\varepsilon}{\sqrt{1 + \alpha^2\varepsilon^2}}\right) = 1 - \frac{\alpha\varepsilon}{\sqrt{1 + \alpha^2\varepsilon^2}}.$$

As $\alpha \to +\infty$,

$$\frac{\alpha\varepsilon}{\sqrt{1 + \alpha^2\varepsilon^2}} = \frac{1}{\sqrt{1 + \dfrac{1}{\alpha^2\varepsilon^2}}} \to 1 - \frac{1}{2\alpha^2\varepsilon^2}.$$

Therefore,

$$\lim_{\alpha \to +\infty}\int_{\varepsilon < |z| < +\infty} \varphi(z)dz \to \frac{1}{2\alpha^2\varepsilon^2} \to 0,$$

for any $\varepsilon > 0$. By Theorem 2.4.1 of *Fourier Analysis* by Stein and Shakarchi, we know

$$\lim_{\alpha\to+\infty} \frac{\alpha}{2(1+\alpha^2 z^2)^{3/2}} = \delta(z).$$

Therefore, the electric field in the limit of $\beta \to 1$, $\gamma \to \infty$ becomes

$$\begin{aligned}
\mathbf{E} &= \lim_{v\to c} \frac{2q}{r_\perp^2} \left(\mathbf{r}_\perp + \gamma(z - vt)\hat{z}\right) \frac{\gamma/r_\perp}{2\left[1 + \left(\dfrac{\gamma}{r_\perp}\right)^2 (z-vt)^2\right]^{3/2}} \\
&= \frac{2q}{r_\perp^2} \left(\mathbf{r}_\perp + \gamma(z - vt)\hat{z}\right) \delta(z - ct) \\
&= \frac{2q}{r_\perp^2} \mathbf{r}_\perp \delta(ct - z).
\end{aligned}$$

For the magnetic field, as $\beta \to 1$, $\boldsymbol{\beta} \to \hat{\mathbf{v}}$, where $\hat{\mathbf{v}}$ is in the z-direction, paralle to the velocity of the particle. Then,

$$\mathbf{B} = \hat{\mathbf{v}} \times \mathbf{E} = 2q \frac{\hat{\mathbf{v}} \times \mathbf{r}_\perp}{r_\perp^2} \delta(ct - z).$$

(b) Using the Maxwell equation, we can show that

$$\rho = \frac{1}{4\pi} \nabla \cdot \mathbf{E} = \frac{q}{2\pi} \nabla_\perp \cdot \left(\frac{\mathbf{r}_\perp}{r_\perp^2}\right) \delta(ct - z).$$

It is straightforward to verify that, for $\mathbf{r}_\perp \neq 0$,

$$\nabla_\perp \cdot \left(\frac{\mathbf{r}_\perp}{r_\perp^2}\right) = 0.$$

On the other hand, for any circle enclosing $\mathbf{r}_\perp = 0$, we have

$$\int \nabla_\perp \cdot \left(\frac{\mathbf{r}_\perp}{r_\perp^2}\right) da = \oint \frac{\mathbf{r}_\perp}{r_\perp^2} \cdot \mathbf{n} dl = \int_0^{2\pi} \frac{\mathbf{r}_\perp}{r_\perp^2} \cdot \mathbf{r}_\perp d\theta = 2\pi.$$

Therefore, symbolically,

$$\nabla_\perp \cdot \left(\frac{\mathbf{r}_\perp}{r_\perp^2}\right) = 2\pi \delta^{(2)}(\mathbf{r}_\perp),$$

and $\rho = q\delta^{(2)}(\mathbf{r}_\perp)\delta(ct - z)$.

For the current,

$$\frac{4\pi}{c} \mathbf{J} = \nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}.$$

By direct calculation, we can show that

$$\begin{aligned}
\nabla \times \mathbf{B} &= \nabla \times \left(2q \frac{\hat{\mathbf{v}} \times \mathbf{r}_\perp}{r_\perp^2} \delta(ct - z)\right) \\
&= 2q \left([\nabla \delta(ct - z)] \times \frac{\hat{\mathbf{v}} \times \mathbf{r}_\perp}{r_\perp^2} + \delta(ct - z) \nabla \times \left[\frac{\hat{\mathbf{v}} \times \mathbf{r}_\perp}{r_\perp^2}\right] \right) \\
&= 2q \left(-\delta'(ct - z) \hat{\mathbf{v}} \times \frac{\hat{\mathbf{v}} \times \mathbf{r}_\perp}{r_\perp^2} + \delta(ct - z) \hat{\mathbf{v}} \left[\nabla \cdot \frac{\mathbf{r}_\perp}{r_\perp^2}\right] \right)
\end{aligned}$$

$$= \quad 2q\left(\frac{\mathbf{r}_\perp}{r_\perp^2}\delta'(ct-z) + 2\pi\hat{\mathbf{v}}\delta^{(2)}(\mathbf{r}_\perp)\delta(ct-z)\right),$$

and

$$\frac{\partial \mathbf{E}}{\partial t} = 2qc\frac{\mathbf{r}_\perp}{r_\perp^2}\delta'(ct-z).$$

Then,

$$\frac{4\pi}{c}\mathbf{J} = 4\pi q\hat{\mathbf{v}}\delta^{(2)}(\mathbf{r}_\perp)\delta(ct-z),$$

or,

$$\mathbf{J} = qc\hat{\mathbf{v}}\delta^{(2)}(\mathbf{r}_\perp)\delta(ct-z).$$

Finally,

$$J^\alpha = (c\rho, \mathbf{J}) = (1, \hat{\mathbf{v}})qc\delta^{(2)}(\mathbf{r}_\perp)\delta(ct-z) = qcv^\alpha\delta^{(2)}(\mathbf{r}_\perp)\delta(ct-z),$$

with $v^\alpha = (1, \hat{\mathbf{v}})$.

(c) From the first gauge, we have

$$\begin{aligned}
\mathbf{E} &= -\nabla\Phi - \frac{1}{c}\frac{\partial \mathbf{A}_1}{\partial t} = -\left(\hat{z}\frac{\partial}{\partial z} + \nabla_\perp\right)A_1^0 - \frac{1}{c}\hat{z}\frac{\partial \mathbf{A}_1^z}{\partial t} \\
&= -2q\hat{z}\delta'(ct-z) + 2q\frac{\mathbf{r}_\perp}{r_\perp^2}\delta(ct-z) + 2q\hat{z}\delta'(ct-z) \\
&= 2q\frac{\mathbf{r}_\perp}{r_\perp^2}\delta(ct-z),
\end{aligned}$$

and

$$\mathbf{B} = \nabla\times\mathbf{A}_1 = \nabla\times(\hat{z}A_1^z) = \nabla A_1^z\times\hat{z} = \nabla_\perp A_1^z\times\hat{z} = -2q\frac{\mathbf{r}_\perp}{r_\perp^2}\delta(ct-z)\times\hat{z} = 2q\frac{\hat{\mathbf{v}}\times\mathbf{r}_\perp}{r_\perp^2}\delta(ct-z).$$

For the second gauge,

$$\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{A}_2}{\partial t} = -\frac{1}{c}\frac{\partial \mathbf{A}_{2,\perp}}{\partial t} = 2q\delta(ct-z)\nabla_\perp\log(\lambda r_\perp) = 2q\frac{\mathbf{r}_\perp}{r_\perp^2}\delta(ct-z),$$

and

$$\begin{aligned}
\mathbf{B} &= \nabla\times\mathbf{A}_2 = \nabla\times\mathbf{A}_{2,\perp} \\
&= \left(\hat{z}\frac{\partial}{\partial z} + \nabla_\perp\right)\times\left(-2q\Theta(ct-z)\nabla_\perp\log(\lambda r_\perp)\right) \\
&= -2q\frac{\partial}{\partial z}\Theta(ct-z)\hat{z}\times\nabla_\perp\log(\lambda r_\perp) \\
&= 2q\delta(ct-z)\hat{z}\times\nabla_\perp\log(\lambda r_\perp) \\
&= 2q\frac{\hat{\mathbf{v}}\times\mathbf{r}_\perp}{r_\perp^2}\delta(ct-z).
\end{aligned}$$

The difference between the two gauges is

$$\Delta A^\alpha = A_1^\alpha - A_2^\alpha = \left(-2q\delta(ct-z)\log(\lambda r_\perp),\ 2q\Theta(ct-z)\nabla_\perp\log(\lambda r_\perp),\ -2q\delta(ct-z)\log(\lambda r_\perp)\right).$$

It is easy to verify that $\Delta A^\alpha = \partial^\mu\chi$, with

$$\chi = -2q\Theta(ct-z)\log(\lambda r_\perp).$$